Session 21- Optional Task

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML
- **Github url:** https://github.com/shrutidarwatkar-sketch/Task_ITR.git

Laptop Price Prediction Using Machine Learning

OUTPUTS

1.Frontends code

```python
app.py > ...
import streamlit as st
import pandas as pd
import joblib


# Load model files
model = joblib.load("Models/Laptop_Model.pkl")
scaler = joblib.load("Models/Scaler.pkl")
columns = joblib.load("Models/Columns.pkl")
label_encoders = joblib.load("Models/LabelEncoders.pkl")

```

```python
st.set_page_config(
    page_title="Laptop Price Prediction",
    page_icon="💻",
    layout="wide"
)

st.title("💻 Laptop Price Prediction")

st.write("Enter laptop details to predict price")


col1, col2 = st.columns(2)


with col1:
    company = st.text_input("Company")
    laptop_type = st.text_input("Type Name")
    ram = st.number_input("RAM (GB)", min_value=2)
    weight = st.number_input("Weight (kg)")


with col2:
    cpu = st.text_input("CPU")
    gpu = st.text_input("GPU")
    storage = st.number_input("Storage (GB)")
    screen = st.number_input("Screen Size")


if st.button("Predict Price"):

    input_data = pd.DataFrame(
        [[company, laptop_type, ram, weight, cpu, gpu, storage, screen]],
        columns=[
            "Company",
            "TypeName",
            "Ram",
            "Weight",
            "Cpu",
            "Gpu",
            "Storage",
            "ScreenSize"
        ]
    )

```

```
)

# Encoding same as training columns
input_data = pd.get_dummies(input_data)

input_data = input_data.reindex(
    columns=columns,
    fill_value=0
)


input_scaled = scaler.transform(input_data)


prediction = model.predict(input_scaled)


st.success(
    f"💰 Estimated Laptop Price: ₹ {round(prediction[0],2)}"
)
```

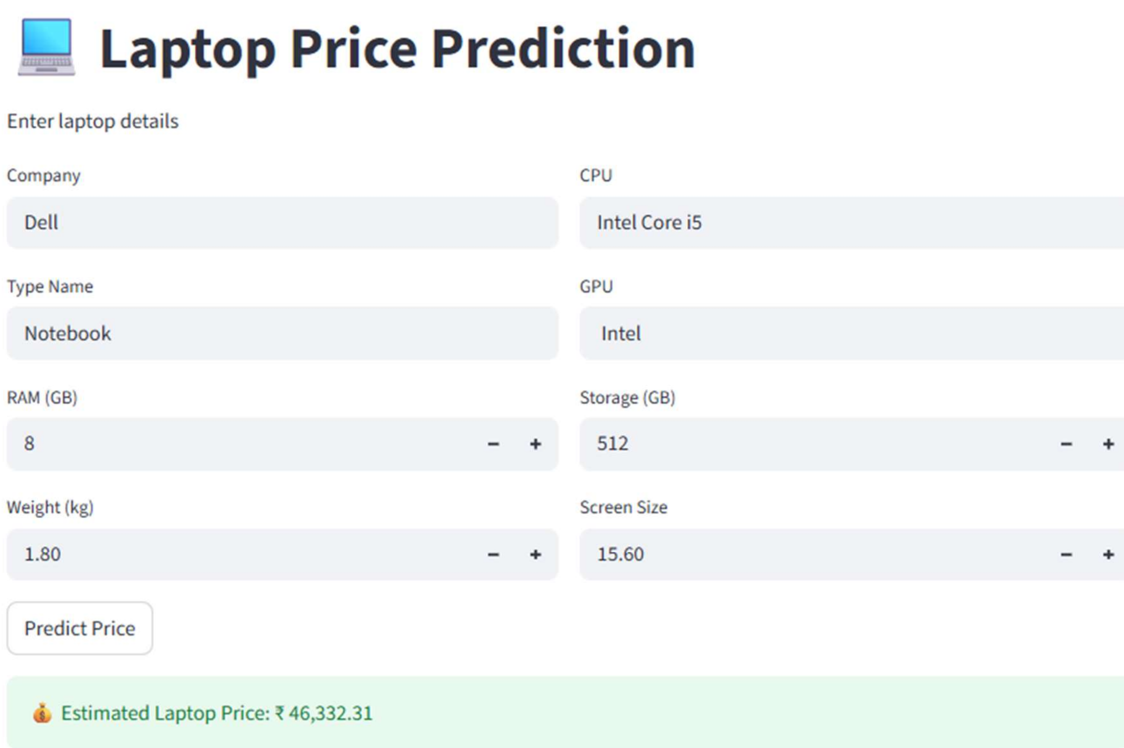
2.App.py output

```
C:\Users\tanve\Desktop\Laptop_Price_Prediction>python -m streamlit run app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.1.11:8501
```

3. Streamlit app home page



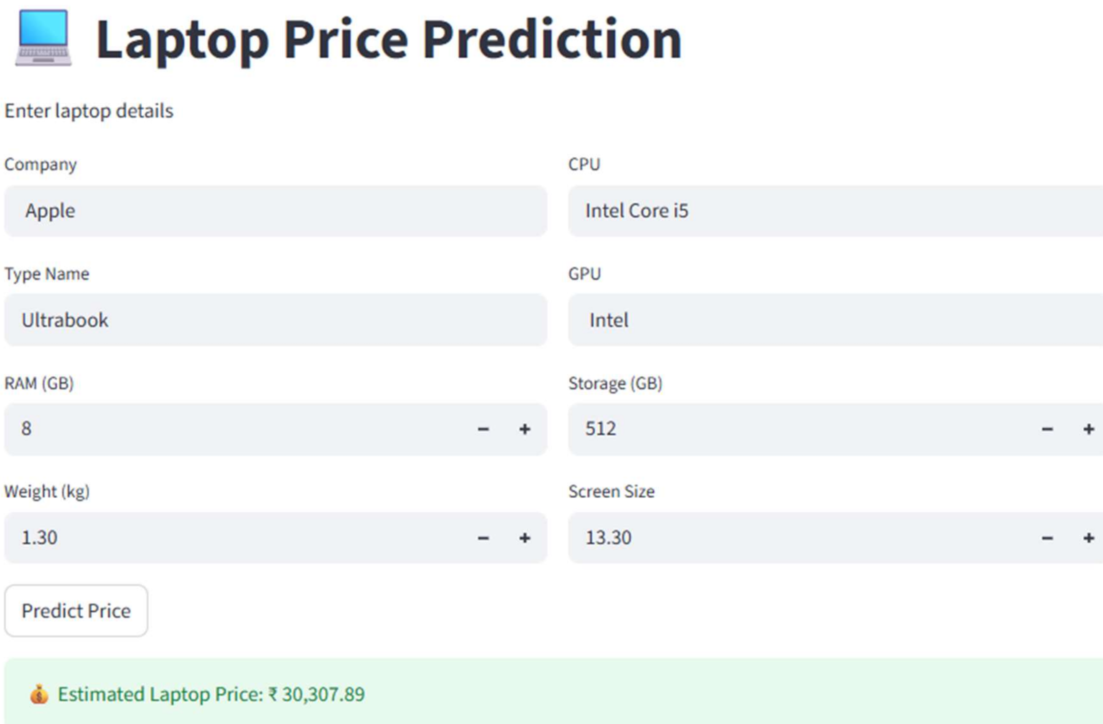
💻 Laptop Price Prediction

Enter laptop details

Company	hp
CPU	Intel Core i7
Type Name	Gaming
GPU	Nvidia
RAM (GB)	16
Storage (GB)	1024
Weight (kg)	2.50
Screen Size	15.60

Predict Price

💰 Estimated Laptop Price: ₹ 28,655.36


Laptop Price Prediction
Enter laptop details
Company
Apple
CPU
Intel Core i5
Type Name
Ultrabook
GPU
Intel
RAM (GB)
8
Storage (GB)
512
Weight (kg)
1.30
Screen Size
13.30
Predict Price
Estimated Laptop Price: ₹ 30,307.89

➢ Conclusion:

- The Laptop Price Prediction project successfully predicts laptop prices based on different specifications.
- The model uses features like company, type, RAM, CPU, GPU, storage, weight, and screen size.
- Data preprocessing and feature encoding were performed before training the model.
- A machine learning model was trained and saved using .pkl files.
- The Streamlit application provides a simple and user-friendly interface for price prediction.
- The project helps users estimate laptop prices quickly and efficiently.
- This project demonstrates the practical use of machine learning in real-world price prediction problems.

➢ **Project Structure**

```
Laptop_Price_Prediction
│
├── app.py
│
├── data
│   └── laptop_data.csv
│
├── Models
│   ├── Laptop_Model.pkl
│   ├── Scaler.pkl
│   └── Columns.pkl
│
└── requirements.txt
```